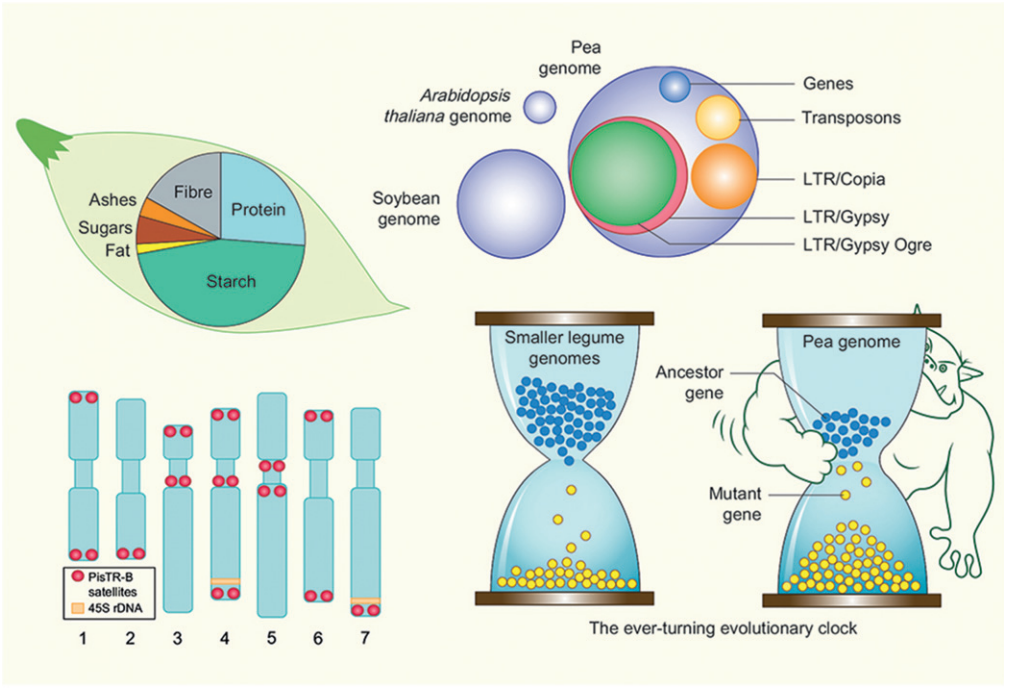
Pisum sativum (Pea)

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Trends in Genetics

The humble pea (*Pisum sativum* L.), which is commonly present at many family tables worldwide, is an important crop for sustainable agriculture and food security, and a famous genetic model, first starring in Mendel's theory of inheritance.

The recently released pea genome sequence is the first pseudo-chromosome level sequence representing the Fabeae, a taxonomic tribe characterised by large genomes. The pea genome contains a large proportion of repeated sequences; most of which are transposable elements (TEs). Since pea's divergence from other tribes, the pea genome has been subject to higher rates of nucleotide mutations and gene duplications and deletions than other sequenced legume genomes. The high abundance of LTR/*Gypsy Ogre* TEs likely influenced the pea genome's rapid evolution. Genome resequencing of *Pisum* accessions further highlighted translocation and transposition events within the genus.

This reference genome is a valuable tool to further explore *Pisum* genetics and evolution, and breed for better cultivars.



Trends in Genetics

TAXONOMY AND CLASSIFICATION:
KINGDOM: Plantae
PHYLUM: Tracheophyta
SUPERCLASS: Angiospermae
CLASS: Magnoliopsida
ORDER: Fabales
FAMILY: Leguminosae (or Fabaceae)
SUBFAMILY: Papillionoidae
TRIBE: Fabeae
GENUS: *Pisum*

GENOME FACTS:
The pea genome spans 4.45 Gb and is organised into seven chromosomes.

The chromosomes have extended primary constrictions due to the presence of multiple domains of centromeric chromatin.

The sequence assembly represents 88% of the genome and contains 44 756 annotated genes and 2 225 175 repetitive elements.

Collapsed sequences of repetitive elements account for most of gap between the genome and the reference assembly.

TEs represent more than 83% of the genome. Long-terminal repeat retrotransposons represent 72.7% of the genome.

The most recent whole genome duplication event in the pea genome is that associated to the crown *Leguminosae* divergence (ca. 55 MYA).

SPECIES FACTS:
Pea seeds are a rich source of gluten-free dietary proteins, as well as fibre, starch, and sugars for human and animal nutrition.

Pea roots host a symbiosis with nitrogen-fixing soil bacteria of the *Rhizobium leguminosarum* species.

By providing an alternative to meat and reducing the use of nitrogen fertiliser, the pea crop contributes to reducing the emission of greenhouse gas.

The *Pisum* genus comprises only two species, *Pisum fulvum* and *Pisum sativum*, the latter including both wild and cultivated subspecies. *P. sativum* subsp. *sativum* was domesticated ca 10 000 years ago in the region of the fertile crescent. *P. sativum* subsp. *abyssinicum*, a cultivated pea type of Ethiopia derived from an independent domestication event.

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Fun fact about the Genome

Pea was Mendel's model for the formulation of the laws of heredity by cross-pollinating pea varieties showing different plant and seed morphological traits. Pea displays a large range of developmental and physiological phenotypes probably thanks to its rapid genome evolution. Pea remained a model for genetics until the advent of molecular biology when its genome size became a serious technical disadvantage. With the recent genome sequence released, pea could well be a great plant model again.

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